

Lecture 3: Output Devices

Monitors (LCD, LED, OLED) · Printers (Inkjet, Laser, 3D) · Plotters · Projectors

Learning Outcomes

- Recall the Input Unit and Input Devices from earlier lectures, and explain the reverse role output devices play.
- Explain, step by step, how LCD and OLED monitors each produce an image, and clarify the relationship between LCD and LED.
- Explain, step by step, how inkjet, laser, and 3D printers each produce their output, and compare all three.
- Explain what a plotter is, describe flatbed and drum plotters, and explain step by step how a plotter draws a line.
- Explain, step by step, how a projector displays an image, and describe the main projector technologies.
- Compare all the output devices in this lecture using exam-ready tables.
- Answer short, medium, and long-answer (up to 15 marks) examination questions on this topic with full step-by-step explanations, diagrams, and examples.

1 Recap — From Input to Output

Lecture 2 studied how real-world actions and objects are converted into binary data by input devices. This lecture studies the reverse journey: how the CPU's binary results are converted back into something a human being can see, hold, or hear. Every output device in this lecture performs that same fundamental job — binary to human-readable form — but each one does it through a completely different mechanism.

Important Point for Exams

As with input devices, examiners rarely just ask you to name an output device. They ask you to explain HOW it works, step by step, or to differentiate between two similar devices (LCD vs OLED, or a flatbed vs drum plotter).

2 What Is an Output Device?

An **output device** is any hardware component that converts the CPU's binary result into a form a human being can understand and use — a visual image, a printed page, a physical model, or a large projected picture. This lecture studies four categories in detail.

Output Devices			
Monitors • LCD, LED, OLED	Printers • Inkjet, Laser, 3D	Plotters • Flatbed, Drum	Projectors • LCD, DLP, Laser

Fig 3.1 — The four categories of output device covered in this lecture.

3 Monitors — LCD, LED & OLED

A monitor (or VDU, Visual Display Unit) presents output as a soft copy — a temporary, on-screen image that disappears once the device is switched off.

Flat Panel Display Technologies		
LCD • Needs a CCFL backlight	LED • An LCD with an LED backlight	OLED • Each pixel emits its own light

Fig 3.2 — LCD, LED, and OLED display technologies.

How an LCD Screen Works — Step by Step

1. A backlight behind the whole panel (traditionally a CCFL, Cold Cathode Fluorescent Lamp) shines light continuously through the display.
2. This light passes through a layer of liquid crystals sandwiched between polarising filters.
3. Applying an electrical voltage to a liquid crystal cell twists the crystals at that point, controlling exactly how much of the backlight is allowed to pass through.
4. Colour filters over each cell split the light into red, green, and blue sub-pixels, which combine to form the full-colour image the viewer sees.

Where Does LED Fit In?

An LED monitor is NOT a separate display technology — it is an LCD monitor that uses an array of LEDs for its backlight instead of older CCFL tubes. Because LEDs are smaller, more energy-efficient, and can be dimmed in zones, LED-backlit LCD monitors are thinner, use less power, and achieve better contrast than CCFL-backlit LCD monitors. This is one of the most frequently misunderstood points in this topic — every LED monitor is fundamentally an LCD monitor at its core.

How an OLED Screen Works — Step by Step

OLED (Organic Light Emitting Diode) is a genuinely different technology from LCD/LED: it needs no backlight at all.

1. An electric current is applied directly to the organic compound layer at each individual pixel.
2. This current causes the organic material to emit its own light directly — a process called electroluminescence — with no separate backlight required anywhere in the panel.
3. The colour and brightness of every individual pixel is controlled independently, simply by varying the current supplied to it.
4. A pixel that should show black is simply given no current at all, so it emits no light whatsoever, producing a true, perfect black.

LCD vs LED vs OLED — Full Comparison

Point	LCD	LED	OLED
Light source	CCFL backlight	LED backlight	Each pixel emits its own light
Black levels / contrast	Lower	Better than CCFL	Perfect blacks
Thickness	Thickest	Thinner	Thinnest; can be flexible
Power for dark content	Same regardless of content	Same regardless of content	Lower — black pixels use no power
Known risk	—	—	Burn-in with static images
Relative cost	Lowest	Moderate	Highest

4 Printers — Inkjet, Laser & 3D

A printer converts the CPU's binary result into a hard copy — a permanent, physical output. Three printer technologies are commonly examined.

Inkjet Printers

An inkjet printer forms text and images by spraying extremely tiny droplets of liquid ink, through microscopic nozzles, directly onto the paper. Most inkjet printers blend cyan, magenta, yellow, and black (CMYK) ink cartridges to produce a full range of colours, which is why they are generally preferred for photographs and colour-rich documents.

How a Laser Printer Works — Step by Step

Fig 3.3 — The five-step laser printing process.

1. Charge: a rotating drum inside the printer is given a uniform electrical charge.
2. Expose: a laser beam scans across the drum, selectively discharging it in the exact pattern of the text or image to be printed.
3. Develop: electrically charged toner powder sticks only to the areas of the drum that the laser discharged.
4. Transfer: the drum rolls over the paper, depositing the toner pattern onto its surface.
5. Fuse: the paper passes through a fuser unit, which applies heat and pressure to permanently melt and bond the toner onto the paper.

How a 3D Printer Works — Step by Step

A 3D printer creates a physical, three-dimensional object by building it up layer by layer, based on a digital 3D model. The most common technology is FDM (Fused Deposition Modeling).

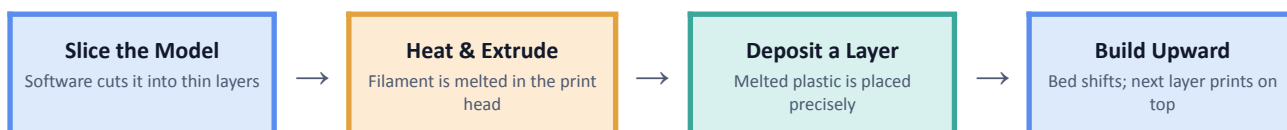


Fig 3.4 — The FDM 3D printing process.

1. A digital 3D model is created or downloaded, then “sliced” by software into hundreds or thousands of thin horizontal layers.
2. A spool of solid plastic filament is fed into a heated print head (extruder), which melts it.
3. The print head moves precisely along the X and Y axes, depositing the melted plastic in the exact pattern of the current layer, which cools and solidifies almost immediately.
4. Once a layer is complete, the print bed moves down by a tiny amount, and the next layer is printed directly on top of it.
5. This process repeats, layer by layer, until the complete three-dimensional object has been built.

Uses: rapid prototyping, custom manufacturing, medical models and prosthetics, education, architectural models.

Inkjet vs Laser vs 3D — Full Comparison

Point	Inkjet	Laser	3D Printing
Output medium	Liquid ink on paper	Toner powder on paper	Solid plastic (or other material)
Result	A flat, 2D image or text	A flat, 2D image or text	A physical, 3D object
Speed	Slower	Much faster	Slowest — can take hours
Best suited for	Photos, colour documents	Bulk text, offices	Prototypes, custom physical parts

5 Plotters

A **plotter** is an output device that produces high-precision, large-format line drawings by moving a pen (or similar tool) across paper, guided by computer instructions. Unlike a printer, which builds an image from a grid of tiny dots, a plotter draws continuous lines directly, making it ideal for technical, line-based drawings.

Type	How It Works
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Flatbed Plotter	The paper stays flat and still on a table-like surface, while a pen mounted on a moving arm draws across it in both the X and Y directions.
Drum Plotter	The paper is wrapped around a rotating drum; the pen moves back and forth along one axis, while the drum itself rotates to provide movement along the other axis.

How a Plotter Draws a Line — Step by Step

1. The computer sends the plotter a series of precise X, Y coordinates describing the lines to be drawn.
2. The pen carriage moves along one axis, while the pen itself (or the bed/drum) moves along the other axis, guided by these coordinates.
3. The pen is lowered onto the paper exactly where a line should be drawn, and lifted when moving to a new position without drawing.
4. This process repeats until the complete technical drawing is finished.

Uses: engineering blueprints, architectural drawings, large maps, circuit board schematics, banners.

6 Projectors

A **projector** takes a video or image signal from a computer and projects an enlarged version of it onto a screen or flat surface, allowing many people to view it at once.

Type	How It Works
LCD Projector	Light shines through three small LCD panels (one each for red, green, and blue), similar in principle to an LCD monitor, and the results are combined through a prism before projecting.
DLP Projector	Uses a chip covered in thousands of microscopic mirrors, each tilting rapidly to reflect light toward or away from the lens to control every individual pixel.
Laser Projector	Uses laser light sources instead of a traditional lamp, offering a longer working life and richer colour.

How a Projector Displays an Image — Step by Step

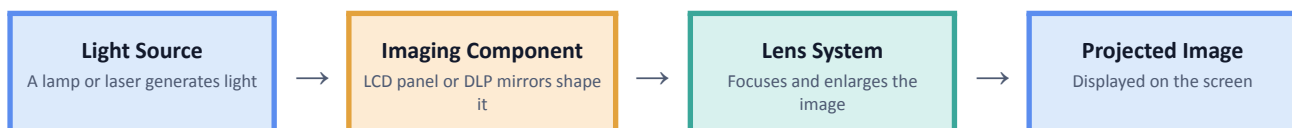


Fig 3.5 — How a projector turns a video signal into a large displayed image.

1. A bright light source — a lamp or a laser — generates the light needed for a large, visible image.
2. This light is shaped into the actual picture, pixel by pixel, by an internal imaging component (LCD panels or a DLP mirror chip), based on the incoming video signal.
3. A system of lenses focuses and enlarges this shaped image.
4. The enlarged image is projected out of the lens onto a screen or wall, where the audience sees it.

7 Full Recap — All Output Devices

Device	Produces	Key Mechanism
LCD Monitor	Soft copy, on-screen image	Liquid crystals control light from a CCFL backlight

LED Monitor	Soft copy, on-screen image	An LCD monitor using LEDs instead of CCFL for backlighting
OLED Monitor	Soft copy, on-screen image	Each pixel emits its own light — no backlight needed
Inkjet Printer	Hard copy, on paper	Liquid ink sprayed through nozzles
Laser Printer	Hard copy, on paper	Toner powder fused onto paper using a charged drum and heat
3D Printer	A physical, 3D object	Melted plastic deposited and built up layer by layer
Flatbed / Drum Plotter	Precise, large-format line drawings	A pen moves across paper along computer-guided coordinates
Projector	A large image on a screen/wall	Light shaped by an imaging component, then focused through a lens

8 Putting It All Together — A Worked Example

Consider an architecture firm designing and presenting a new building to a client:

Step	Device Used	What Happens
1	OLED Monitor	The architect designs the building's 3D model on a high-contrast OLED monitor, which shows deep shadows and rich detail clearly.
2	Laser Printer	Written specification documents are printed quickly and cheaply in bulk using a laser printer.
3	Plotter	The precise architectural blueprint, showing exact structural dimensions, is printed at large format on a plotter.
4	3D Printer	A small physical scale model of the building is created using a 3D printer, to show the client.
5	Projector	During the client meeting, the finished design is displayed on a large screen using a projector, so everyone in the room can see it clearly.

This single real-world scenario uses every output device in this lecture, each chosen specifically for the job it does best: a screen for detailed design work, a printer for bulk documents, a plotter for precise technical drawings, a 3D printer for a physical model, and a projector for group presentation.

9 Fill in the Blanks

- 1. An LED monitor is actually a type of _____ display that uses LEDs for backlighting.
- 2. In a(n) _____ display, each individual pixel emits its own light, with no backlight required.
- 3. The five-step laser printing process is Charge, Expose, Develop, Transfer, and _____.
- 4. The most common 3D printing technology, which melts and deposits plastic filament layer by layer, is called _____.
- 5. Unlike a printer, a(n) _____ draws continuous lines directly using a pen, rather than building an image from tiny dots.

- 6. A _____ plotter keeps the paper flat and still, while the pen moves across it in both directions.
- 7. A projector that uses thousands of microscopic tilting mirrors to control each pixel is called a(n) _____ projector.
- 8. Because a black pixel on an OLED screen uses no power at all, OLED displays are especially efficient when showing _____ content.
- 9. The risk of a static image becoming permanently visible on an OLED screen is called _____.
- 10. In 3D printing, the process of cutting a digital model into thin horizontal layers is called _____.

Answers

1. LCD (Liquid Crystal Display) | 2. OLED | 3. Fuse | 4. FDM (Fused Deposition Modeling) | 5. Plotter | 6. Flatbed | 7. DLP (Digital Light Processing) | 8. Dark | 9. Burn-in | 10. Slicing

10 True or False

- 1. An OLED display requires a separate backlight, just like an LCD.
- 2. A laser printer uses dry toner powder rather than liquid ink.
- 3. 3D printing builds an object layer by layer from a digital model.
- 4. A plotter is better suited than a standard printer for producing precise, large-format technical line drawings.
- 5. A DLP projector uses three small LCD panels, one for each primary colour.
- 6. OLED displays can achieve perfect blacks because a black pixel simply receives no current.
- 7. A flatbed plotter wraps the paper around a rotating drum.
- 8. Laser printers are generally faster and cheaper per page than inkjet printers for bulk text printing.

Answers with Explanation

- 1. False.** An OLED pixel emits its own light directly; it needs no separate backlight, unlike LCD or LED.
- 2. True.** Laser printers use toner powder, fused with heat; liquid ink is used by inkjet printers.
- 3. True.** That is precisely how 3D printing works, building up an object one thin layer at a time.
- 4. True.** A plotter draws continuous, highly precise lines directly, ideal for technical drawings, unlike a standard printer's dot-based image.
- 5. False.** That describes an LCD projector; a DLP projector uses a chip of microscopic tilting mirrors instead.
- 6. True.** Since no current is applied, no light is emitted at all, producing a true, perfect black.
- 7. False.** It is the reverse: a drum plotter wraps the paper around a rotating drum, while a flatbed plotter keeps the paper flat and still.
- 8. True.** Laser printers are faster and cheaper per page for bulk text printing.

11 Short Answer Questions (2–3 Marks)

- Differentiate between an LCD and an OLED display. (3 marks)
- Why is an LED monitor not considered a separate technology from an LCD monitor? (2 marks)
- Name the five steps of the laser printing process, in order. (2 marks)
- What is 3D printing? Name one common use. (2 marks)
- Differentiate between a flatbed plotter and a drum plotter. (3 marks)

- Why is a plotter better suited than a printer for engineering drawings? (2 marks)
- Name three types of projector technology. (2 marks)
- What is FDM in 3D printing? (2 marks)
- Why do OLED displays use less power for dark content? (2 marks)

12 Long Answer Model Answers

The following are fully written model answers. Use these as a template for structure, depth, and the level of explanation expected in your own exam answer — do not simply memorise them word for word; practise writing them in your own words using the same structure.

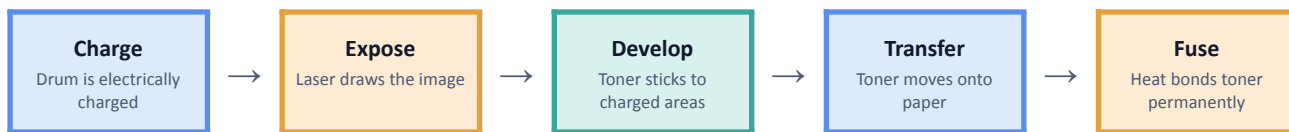
Q1 (5 Marks): Differentiate between LCD, LED, and OLED monitors.

An LCD monitor is built from a layer of liquid crystals sandwiched between polarising filters, illuminated from behind by a backlight — traditionally a CCFL tube. Applying voltage to each cell twists its crystals to control how much of that backlight passes through, and colour filters produce the final image.

An LED monitor is not a separate technology — it is an LCD monitor that uses an array of LEDs for its backlight instead of CCFL tubes, making it thinner, more energy-efficient, and capable of better contrast.

An OLED monitor works completely differently: each individual pixel is made of an organic compound that emits its own light directly when a current is applied, so no backlight is needed at all. Because a black pixel receives no current and emits no light, OLED achieves perfect blacks and very high contrast, though it is more expensive and can suffer from burn-in with static images.

Q2 (10 Marks): Explain, step by step, the working of a laser printer and a 3D printer, and compare them.



(The five-step laser printing process, as in Fig 3.3 of these notes.)

A laser printer works through five steps: the drum is Charged uniformly; a laser beam Exposes it, discharging the exact image pattern; charged toner powder Develops onto the discharged areas; the drum Transfers this toner onto the paper; and finally the paper passes through a Fuser, which uses heat and pressure to permanently bond the toner.

A 3D printer works completely differently, building a physical object rather than a flat image. Software first Slices a digital 3D model into thin horizontal layers. A print head then melts plastic filament and, moving precisely along the X and Y axes, Deposits it in the pattern of the current layer. Once a layer is complete, the print bed shifts down slightly, and the process repeats, Building the object upward, layer by layer, until it is complete.

Point	Laser Printer	3D Printer
Output	A flat, 2D image or text on paper	A physical, 3D object
Material	Toner powder	Plastic filament (or other material)
Speed	Fast, seconds per page	Slow, can take hours

In conclusion, both printers build their output through a fixed, repeatable physical process, but a laser printer's process ends with a flat page, while a 3D printer's process only ends once an entire three-dimensional shape has been constructed.

Q3 (10 Marks): Explain what a plotter is, describe its types, and explain step by step how it produces a drawing.

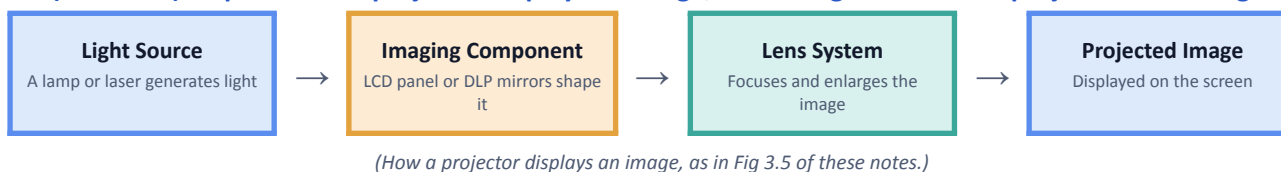
A plotter is an output device that produces high-precision, large-format line drawings by moving a pen across paper, guided by computer instructions. Unlike a printer, which builds an image from a grid of tiny dots, a plotter draws continuous lines directly, which is why it is preferred for technical, line-based drawings such as engineering blueprints and architectural plans.

Type	How It Works
Flatbed Plotter	The paper stays flat and still, while a pen mounted on a moving arm draws across it in both the X and Y directions.
Drum Plotter	The paper is wrapped around a rotating drum; the pen moves along one axis while the drum rotates to provide movement along the other.

1. The computer sends the plotter a series of precise X, Y coordinates describing the lines to be drawn.
2. The pen carriage moves along one axis, while the pen itself (or the bed/drum) moves along the other, guided by these coordinates.
3. The pen is lowered onto the paper exactly where a line should be drawn, and lifted when moving without drawing.
4. This process repeats until the complete technical drawing is finished.

In conclusion, a plotter's advantage over a standard printer lies precisely in this pen-based, coordinate-driven method: it produces sharp, continuous, perfectly scaled lines of any length, which is essential for technical drawings where dot-based printing would look jagged or imprecise.

Q4 (10 Marks): Explain how a projector displays an image, describing the different projector technologies.



A projector takes a video signal from a computer and projects an enlarged image onto a screen. First, a bright Light Source — a lamp or a laser — generates the light needed for a large, visible image. Second, an internal Imaging Component shapes this light into the actual picture, pixel by pixel. Third, a Lens System focuses and enlarges this shaped image. Finally, the enlarged image is projected onto a screen or wall as the final visible picture.

Type	How the Imaging Component Works
LCD Projector	Light shines through three small LCD panels (red, green, blue), combined through a prism.
DLP Projector	A chip of thousands of microscopic mirrors tilts rapidly to control each pixel.
Laser Projector	Uses laser light sources instead of a lamp, for longer life and richer colour.

In conclusion, all three projector types follow the same overall four-stage process — generate light, shape it into an image, focus it, and project it — differing mainly in how the second stage, shaping the light into a picture, is actually achieved.

Q5 (15 Marks): Describe, with a suitable example, how an organisation such as an architecture firm could use multiple output devices covered in this lecture together, explaining the role of each.

Consider an architecture firm designing and presenting a new building to a client, which draws on almost every output device studied in this lecture, each solving a different part of the same overall task.

Step	Device Used	Role
1	OLED Monitor	Used for detailed design work, since its perfect blacks and high contrast make fine shadow and material detail easy to see accurately.

2	Laser Printer	Prints written specification documents quickly and cheaply in bulk.
3	Plotter	Produces the precise, large-format architectural blueprint, with exact structural dimensions drawn as sharp, continuous lines.
4	3D Printer	Builds a small physical scale model of the building, giving the client something tangible to examine.
5	Projector	Displays the finished design on a large screen during the client meeting, so everyone in the room can see it clearly at once.

Each device is chosen specifically for the strength it offers: the OLED monitor for accurate, high-contrast design work; the laser printer for fast, cheap bulk documents; the plotter for precise technical drawings a standard printer cannot match; the 3D printer for a physical model no 2D output could provide; and the projector for showing the result to a whole room of people at once.

In conclusion, no single output device could serve every stage of this process alone; it is the combination of devices — each converting the same underlying digital design into a different physical or visual form — that allows a firm to design, document, model, and present its work effectively.

13 Exam Tips

- Memory trick for monitors: LCD needs a backlight; LED is an LCD with an LED backlight; OLED needs NO backlight at all — each pixel is its own light source.
- Memory trick for the laser printing process: “Charge, Expose, Develop, Transfer, Fuse” — the order is always tested.
- Memory trick for 3D printing: “Slice, Heat & Extrude, Deposit, Build Upward” — think of it as printing one thin 2D layer at a time, stacked into a 3D shape.
- Memory trick for plotters: Flatbed = paper stays FLAT, pen moves both ways. Drum = paper wraps the DRUM, which rotates.
- For “explain the working of...” questions (LCD, OLED, laser printer, 3D printer, plotter, projector), always answer as a NUMBERED, step-by-step sequence — exactly like the model answers in Section 12.
- For “differentiate” questions (LCD vs OLED, flatbed vs drum plotter, LCD vs DLP projector), always answer using a table — it is faster to write and easier for the examiner to award marks clearly.
- Common Mistake: Do not describe LED monitors as a separate technology from LCD — an LED monitor is an LCD monitor with an LED backlight.
- Common Mistake: Do not say laser printers use “ink” — they use dry toner powder. Only inkjet printers use liquid ink.
- For 10 and 15 mark questions, structure your answer with a clear paragraph or table per device, exactly like the model answers in Section 12 — examiners reward clear structure over long, unstructured paragraphs.

References

- Refer to your university-prescribed/standard Computer Fundamentals textbook for this topic, as per the syllabus official reading list.
- Class lecture discussion and board work for this lecture.
- MDC Section II, Lecture 2 — Input Devices, for the companion topic this lecture builds on.